PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · BIOLOGY · 4BI1/2B

A* Examination Guide

Built from 17 mark schemes and 16 question papers, January 2019 to June 2026

This guide is built from the Paper 2B question papers and mark schemes. Model answers follow the marking points examiners list, and every warning comes from an “ignore”, “reject”, “allow” or “no credit” note. Paper 2 examines the whole specification including the bold Paper-2-only content: protein synthesis, menstrual-cycle hormones, the nitrogen cycle, transpiration, the kidney, immunity and biotechnology. Most questions are set in an unfamiliar context with data.

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Papers analysed: Jan 2019 (legacy 4BI0, mark scheme only), Jun 2019, Jan 2020, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 How Paper 2B is built

Format: 1 hour 15 minutes, 70 marks (the legacy 4BI0 paper was 1 hour, 60 marks), 6–7 longer questions, each built around a context: a disease outbreak, a contraceptive trial, a reef survey, a transplant programme. About 1 minute per mark. Expect calculations in standard form, a 4–6 mark method, a 4–6 mark data evaluation and a 6-mark extended explanation.

Topic	Papers (of 17)	What is asked most
Ecology, cycles, pollution	14	greenhouse gases, carbon/nitrogen cycles, eutrophication, quadrats
Genetics and protein synthesis	12	transcription/translation, codominance, sickle cell, crosses
Immunity and disease	11	vaccination, phagocytes/lymphocytes, antibiotic resistance, transplants
Plant physiology	11	photosynthesis and light colour, transpiration and potometers, stomata
Biotechnology	10	GM (restriction enzyme, ligase), cloning, fermenters, fish farming, stem cells
Reproductive hormones	7	FSH, LH, oestrogen, progesterone; contraception and fertility
Kidney and homeostasis	7	dialysis, ADH, transplants, sweating
Enzyme investigations	6	catalase and potato discs, decomposition, temperature and pH
Natural selection and breeding	6	mutation → survival → reproduction; selective breeding; polygenic traits

2 TWENTY MARK-SCHEME RULES

1. **Standard form:** give $a \times 10^n$ with $1 \le a < 10$; 18×10^{12} or the right digits with the wrong power earn partial credit only.
2. **"Ignore amount", "ignore nutrients", "ignore food":** name the substance (glucose, amino acids, oxygen) and the quantity.
3. **Balanced symbol equations score;** word equations score 0 when the equation is asked ($6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$).
4. **Translation wording:** the tRNA **anticodon** binds to the mRNA **codon**; reversing them is rejected.
5. **Codominance:** "both alleles are expressed / affect the phenotype"; "both genes are expressed" is rejected.
6. **Natural selection needs the full chain:** mutation → variation → survive → reproduce → pass on the allele.
7. **Immunity roles:** phagocytes engulf; lymphocytes make antibodies. "Lymphocytes engulf" and "phagocytes produce antibodies" are rejected; "fight" and "attack" are ignored.
8. **Vaccination needs:** attenuated/dead pathogen with antigens, memory cells, faster and larger secondary antibody response.
9. **Antibiotic resistance:** mutation, resistant bacteria survive, reproduce, pass on the allele; "immune" is ignored.
10. **Greenhouse effect:** traps infrared radiation/heat → global warming; "traps light" is ignored, "carbon emissions" unqualified ignored.
11. **Mineral ions with matched functions:** magnesium → chlorophyll, nitrate → amino acids/proteins; "nitrogen" and "phosphorus" are ignored.
12. **Restriction enzyme and ligase:** cut and join; "restricted", "lipase" are rejected.
13. **Osmosis in water potential terms;** salt does not move by osmosis.
14. **Stem cells:** undifferentiated and can divide by mitosis; meiosis is rejected.
15. **Methods (4–6):** same organism/size, change one variable, measure with named apparatus after a set time, repeats and means, named control variables.
16. **Evaluations:** for and against with data; sample size, duration, side effects, controls, and whether the data support the claim.
17. **Qualitative measurements** (cloudiness, colour) are subjective; suggest a quantitative method.
18. **Means:** exclude anomalies ($4.2 \div 3$, not $\div 4$) and keep the precision of the data.
19. **Prevent evaporation** (oil layer on a potometer reservoir) so water loss is only from the plant.
20. **Context matters:** link every point to the organism in the question; generic textbook answers often score only one mark.

3 ENZYMES AND INVESTIGATIONS

CATALASE AND POTATO DISCS (JAN 21, NOV 23)

More discs (more enzyme) → more active sites → more enzyme–substrate complexes → more oxygen; measure gas with a syringe or inverted measuring cylinder. Measure the initial rate because the substrate is used up as time passes.

Control: volume and concentration of hydrogen peroxide, pH buffer, temperature, size/surface area of discs. Cutting into smaller pieces increases surface area so more catalase is exposed.

- Decomposition (Jan 22): fungi/bacteria release enzymes for extracellular digestion; cold slows enzymes and respiration; vinegar (low pH) denatures enzymes, so food keeps longer.
- Urease and pH (Jun 21): away from the optimum the active site changes shape, fewer enzyme–substrate complexes form.

- Food energy method (Jan 23): weigh the food, measure a set volume of water and its starting temperature, burn completely under the tube, record the highest temperature, repeat; energy = mass of water × 4.2 × ΔT; heat is lost to the air, so values are low.

4 PHOTOSYNTHESIS AND GAS EXCHANGE IN PLANTS

LIGHT COLOUR (NOV 20)

No filter gives the highest rate (all wavelengths); red and blue light are absorbed by chlorophyll, so the rate is high; green is reflected or transmitted, so the rate is lowest. Counting bubbles is inaccurate (different sizes, oxygen dissolves, bubbles may contain other gases); measure volume with a syringe or use hydrogencarbonate indicator.

- Hydrogencarbonate indicator (Nov 23): in light, photosynthesis > respiration, CO_2 falls, indicator turns purple; in the dark it turns yellow (more CO_2). Dim light: compensation point, no change (orange-red).
- Leaf structure (Jun 19): palisade cells near the top, closely packed with many chloroplasts to absorb light; spongy layer has air spaces for gas exchange; most stomata on the lower surface.
- Starch test: boil, warm ethanol in a water bath, rinse, iodine → blue-black.
- Carbon dioxide is a raw material and can be a limiting factor (Nov 23); carbon sinks absorb more CO_2 than they release.

5 TRANSPIRATION AND PLANT TRANSPORT

POTOMETER METHOD (5–7) · JUN 24

Cut the shoot under water (diagonally), fit it into the potometer, seal joints with petroleum jelly, dry the leaves; measure the distance the bubble moves in a set time; change light intensity by moving the lamp; keep temperature, humidity and air movement the same; reset with the reservoir and repeat.

- Light: more stomata open (until all are open) → more transpiration. Desert plants close stomata in high light and have fewer stomata.
- Wind (Jan 20, Jun 23): blows away humid air, maintains the concentration gradient, more diffusion/evaporation.
- Transpiration draws water up the xylem from the roots, cools the plant, carries minerals, keeps cells turgid.
- Phloem carries sucrose and amino acids (Nov 25 aphids).
- Stomatal density (Nov 23): 2 stomata in 0.04 × 0.04 cm = 0.0016 cm^2 ⇒ 1250 per cm^2 ⇒ 150 cm^2 leaf has 187 500 stomata.
- Mass vs volume of water (Jun 23): uptake exceeds loss because some water is used in photosynthesis and for turgor.

6 EXCRETION AND HOMEOSTASIS

MODEL ANSWER · WATER BALANCE AFTER EXERCISE (3–5) · JUN 24

Sweating cools the body by evaporation; blood becomes more concentrated; the hypothalamus detects this; the pituitary releases ADH; the collecting duct becomes more permeable; more water is reabsorbed; less, more concentrated urine. Pure water dilutes the blood, so less ADH and more urine; isotonic drinks also replace salts and sugar.

DIALYSIS AND TRANSPLANT (JAN 23, NOV 24)

Dialysis design: dialysis fluid has no urea (diffusion gradient), long coiled tubing (large surface area), thin membrane (short diffusion distance), counter-flow of fluid (maintains gradient), 40 °C (faster diffusion).

Transplant: permanent, normal diet and life, no hospital visits; donor must be a close tissue match (similar antigens); immunosuppressant drugs prevent rejection.

- Homeostasis: maintaining a constant internal environment. Urine contains urea, water and salts.
- Heart (Nov 21): atria receive blood and pump it into the ventricles; veins have lower pressure, less oxygen, more CO_2.
- Heart–lung machine oxygenates blood and removes CO_2; smoking damages arteries and CO binds haemoglobin.

7 REPRODUCTIVE HORMONES

Hormone	Secreted by	Functions
FSH	pituitary	follicle (egg) maturation; stimulates oestrogen
LH	pituitary	ovulation; corpus luteum; progesterone release
oestrogen	ovaries	repairs uterus lining; stimulates LH
progesterone	ovaries / corpus luteum / placenta	maintains the lining; inhibits FSH and LH

- Menstruation: breakdown and loss of the uterus lining about once a month.
- Eggs released in a lifetime (Nov 20): $(51 - 12) \times 13$ cycles $= 507 \approx 510$.
- Male contraceptive trial (Jun 21): progestin lowers testosterone and sperm production; evaluate effectiveness (98.4%), side effects, reversibility, time to work, no protection against STIs.
- Phyto-oestrogens (Nov 25): thicken the lining for implantation; high oestrogen stimulates LH and ovulation.
- Low FSH ⇒ eggs do not mature; low LH ⇒ no ovulation (Jun 25).

8 GENETICS AND PROTEIN SYNTHESIS

PROTEIN SYNTHESIS (4–6) · JAN 21, JUN 26

Transcription: DNA is copied into mRNA in the nucleus. Translation: mRNA moves to a ribosome; the ribosome moves along the mRNA codons; tRNA brings specific amino acids; the tRNA anticodon binds to the complementary mRNA codon; amino acids are joined into a polypeptide. Antibiotics that stop ribosomes binding prevent translation (Nov 20).

- Codominance (Jan 22): blood groups I^AI^B; O has no antigens, so O is the universal donor.
- Sickle cell (Jun 26): lower surface area, less oxygen carried, less respiration; carriers survive malaria and pass on the allele.
- Polygenic traits (height) are controlled by several genes and the environment; monogenic traits (hair colour) by one gene.
- Crosses: show genotypes, gametes, offspring; P = 0.25 for a recessive child from two carriers (Jan 23).
- Gene editing (Jun 26): guide RNA may bind similar sequences elsewhere; ethical issues, unknown effects, future generations cannot consent.

9 NATURAL SELECTION AND BREEDING

NATURAL SELECTION (4–5) · JUN 23, NOV 25

Mutation produces variation (glowing dinoflagellates; plants with terpenes); individuals with the useful feature survive (are not eaten), reproduce and pass on the allele; over generations the allele becomes more common.

- Selective breeding (Jan 23): choose the tallest cattle, breed them, select the tallest offspring, repeat over many generations.
- Antibiotic resistance follows the same chain; MRSA is the classic example.

10 IMMUNITY AND DISEASE

- White blood cells: phagocytes engulf pathogens (phagocytosis); lymphocytes produce specific antibodies and memory cells.
- Vaccination (Jan 19, Jan 20): attenuated or dead pathogen, antigens trigger antibodies and memory cells; secondary response is faster and larger.
- Virus vs bacterium: smaller, protein coat, no cell wall, membrane, cytoplasm or ribosomes; contains DNA or RNA.
- Vectors and climate (Jan 19): warmer, wetter, more humid conditions help mosquitoes breed; standing water is a breeding site.
- Bovine TB (Jan 20): pasteurisation kills bacteria; testing a badger vaccine needs two comparable areas with and without vaccination.
- Transplants: tissue-matched donors and immunosuppressants prevent rejection (antigens recognised as foreign).

11 ECOLOGY, CYCLES AND POLLUTION

EUTROPHICATION (4) · JUN 25

Fertiliser or sewage (nitrates, phosphates) runs off into water; algae grow rapidly (algal bloom), blocking light; plants below cannot photosynthesise and die; decomposers respire, using oxygen; fish and other animals die from lack of oxygen.

NITROGEN CYCLE (5) · JUN 21, JUN 22

Nitrogen-fixing bacteria: N_2 → ammonium/nitrates; decomposers: proteins → ammonium; nitrifying bacteria: ammonium → nitrite → nitrate; denitrifying bacteria: nitrate → N_2 (in waterlogged soil). Plants absorb nitrate by active transport to make amino acids.

- Ecosystem: all the organisms in an area and their environment. Quadrats placed randomly with random-number coordinates.
- Reef protection (Jun 19): reduce greenhouse gases, limit tourism and fishing, replant coral.
- Global warming effects (Jun 22): melting ice, sea-level rise, habitat loss, extinctions, disease spread, extreme weather.
- Deforestation (Jun 26): fewer roots holding soil, more erosion, less transpiration, less CO_2 absorbed.
- Microplastics (Jan 21): biodegradable plastics are broken down by microorganisms; less harm to animals.

12 BIOTECHNOLOGY

- Genetic engineering: restriction enzyme cuts the gene and plasmid (sticky ends); DNA ligase joins them; vector inserts the recombinant DNA.
- Micropropagation/cloning (Jun 22, Jan 23): large numbers quickly, all genetically identical with the desired gene, any time of year.
- Fermenters (Jan 20): paddles mix microorganisms with nutrients and oxygen and keep temperature even; water jacket cools; aseptic conditions.
- Fish farming (Jun 21): filtered water, more oxygen, less disease, faster growth; nets keep predators out.
- Stem cells (Jan 22): undifferentiated, divide by mitosis and differentiate; artificial red cells lack the biconcave shape (less oxygen uptake).
- Pest control (Nov 25): compare pesticides with data on aphid numbers over time; resistance and timing.

13 CALCULATIONS

Type	Past-paper example
percentage	43 ÷ 1300 × 100 = 3.3% (Jan 19); 178 500 ÷ 180 000 × 100 = 99.2% (Nov 21)
percentage change	(8.3 − 7.3) ÷ 7.3 × 100 = 13.7% (Jan 21); (2.4 − 0.8) ÷ 0.8 × 100 = 200% (Nov 23)
percentage of a number	1.6% of 320 ≈ 5 (Jun 21); 75% of 200 = 150 (Nov 21)
standard form	4.74×10^7 (Jan 22); 1.8×10^{13} (Jun 22); 9.5×10^8 (Jan 23)
ratio	147 710 : 13 297 = 11.1 : 1 (Jan 20)
rate	10 kg ÷ 180 days = 0.056 kg/day (Jun 21)
scaling	14.6 × 60 × 24 = 21 024 (Jan 19); 1250 per cm^2 × 150 = 187 500 stomata (Nov 23)
mean	(84 + 80) ÷ 2 = 82 (Nov 20); 4.2 ÷ 3 = 1.4, excluding the anomaly (Jan 21)

14 FINAL A* CHECKLIST

1. Use the context: name the organism, process and data in every point.
2. Protein synthesis: anticodon on tRNA binds codon on mRNA at the ribosome.
3. Immunity: phagocytes engulf, lymphocytes make antibodies and memory cells.
4. Natural selection and resistance: mutation → survive → reproduce → pass on allele.
5. Hormones: FSH (follicle), LH (ovulation), oestrogen (repairs), progesterone (maintains).
6. Cycles and eutrophication: name each bacterium and each conversion; oxygen depletion by decomposers.
7. Calculations: standard form, units, the s.f. asked; exclude anomalies from means.
8. Methods and evaluations: CORMS-style detail; points for and against. Leave 5 minutes to check.